

Exercise-2

Basics and Operators

a) Sum and Difference of 2 numbers Objective: Learn int and float data types.

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b;
```

```
    float x, y;
```

```
    scanf("%d %d", &a, &b);
```

```
    scanf("%f %f", &x, &y);
```

```
    printf("Integer Sum = %d\n", a + b);
```

```
    printf("Integer Difference = %d\n", a - b);
```

```
    printf("Float Sum = %.2f\n", x + y);
```

```
    printf("Float Difference = %.2f\n", x - y);
```

```
    return 0;
```

```
}
```

O/P

50 20

45.8 22.6

Integer Sum = 70

Integer Difference = 30

Float Sum = 68.40

Float Difference = 23.20

b) Playing with Characters Objective: Learn how to take a character, a string and a sentence as input in C.

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    char ch;
```

```
    char str[50];
```

```
    char sentence[100];
```

```
    scanf("%c", &ch);    // Read a character
```

```
    scanf("%s", str);    // Read a string (word)
```

```
    scanf(" %[^\n]", sentence); // Read a full sentence
```

```
    printf("%c\n", ch);
```

```
    printf("%s\n", str);
```

```
printf("%s", sentence);
```

```
return 0;
```

```
}
```

O/P

r

ramya

ramya Chikkireddi

r

ramya

ramya Chikkireddi

C) Bitwise Operators Objective: Learn how to work with bits (0,1) and bitwise operators.

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
int a, b;
```

```
scanf("%d %d", &a, &b);
```

```
printf("a & b = %d\n", a & b);
```

```
printf("a | b = %d\n", a | b);
```

```

printf("a ^ b = %d\n", a ^ b);
printf("~a = %d\n", ~a);
printf("a << 1 = %d\n", a << 1);
printf("a >> 1 = %d\n", a >> 1);

return 0;
}

```

O/P

12 10

$a \& b = 8$

$a | b = 14$

$a \wedge b = 6$

$\sim a = -13$

$a \ll 1 = 24$

$a \gg 1 = 6$

D) Conversion of Fahrenheit to Celsius and vice versa.

Temperature Conversion Formula
Fahrenheit to Celsius
$C = \frac{5}{9}(F - 32)$

//Conversion of Fahrenheit to Celsius

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    float f, c;
```

```
    scanf("%f", &f);
```

```
    c = (f - 32) * 5 / 9;
```

```
    printf("%.2f", c);
```

```
    return 0;
```

```
}
```

O/P

100

37.78

Temperature Conversion Formula
Celsius to Fahrenheit
$F = \frac{9}{5}C + 32$

//Conversion of Celsius to Fahrenheit

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    float c, f;
```

```
scanf("%f", &c);  
f = (c * 9 / 5) + 32;  
printf("%.2f", f);  
return 0;  
}
```

O/P

0

32.00

e) Distance travelled by an object.

formula:

$$\text{Distance} = \text{Speed} \times \text{Time}$$

```
#include<stdio.h>  
  
int main()  
{  
    float speed, time, distance;  
    scanf("%f", &speed);  
    scanf("%f", &time);  
    distance = speed * time;  
    printf("%.2f", distance);  
    return 0;  
}
```

O/P

60

2

120.00

F) Calculate Simple interest and compound interest.

Formula for Simple Interest

$$SI = (P \times R \times T) / 100$$

Where:

- **P = Principal amount**
- **R = Rate of interest**
- **T = Time**

// Simple interest

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
float P, R, T, SI;
```

```
scanf("%f", &P);
```

```
scanf("%f", &R);
scanf("%f", &T);
SI = (P * R * T) / 100;
printf("%.2f", SI);
return 0;
}
```

O/P

10000

2

20

4000.00

Compound Interest
Formula:
$A = P \left(1 + \frac{R}{100}\right)^T$ $CI = A - P$

Where:

- **P** = Principal
- **R** = Rate of Interest
- **T** = Time
- **A** = Total Amount

- **CI** = Compound Interest

//Compound Interest

```
#include<stdio.h>
```

```
#include<math.h>
```

```
int main()
```

```
{
```

```
    float p, r, t, ci,a;
```

```
    scanf("%f %f %f", &p, &r, &t);
```

```
    a = p * pow((1 + r / 100), t) ;
```

```
    ci=a-p;
```

```
    printf("%.2f", ci);
```

```
    return 0;
```

```
}
```

O/P

1500

10

2

315.00